

LEARNQUEST

LINUX LABS DOCUMENTATION

MODULES 1 & 2

Navigation, File Basics & Process Management

Command-Line Practice Documentation

Linux Fundamentals Coursework

July 2026

1. Introduction

Working confidently in Linux begins with knowing where you are in the filesystem, what files and directories surround you, and how to read, copy, move, and remove them. Beyond files, every running Linux system is also a collection of processes — some tied to what a user is doing directly, and others, called daemons, running quietly in the background to provide services such as web servers or databases.

This document brings together the two modules of hands-on lab practice completed so far. Module 1 covers filesystem navigation, listing, viewing, copying, and removing files. Module 2 introduces process management — inspecting what is running on a system and terminating a process when needed. Commands are grouped by function rather than listed in one long sequence, with each category followed by the terminal screenshots that demonstrate it in action.

2. Learning Objectives

By the end of these modules, the learner will be able to:

- Navigate the Linux filesystem using `pwd` and `cd`.
- List files and directories in multiple formats, including hidden files, using `ls` and its options.
- View and copy the contents of text files using `cat` and `cp`.
- Remove files and directories safely with `rm` and `rm -r`, and rename or relocate them with `mv`.
- Explain what a daemon is and recognize background services running on a system.
- Inspect running processes and their resource usage using `ps -aux`.
- Terminate a specific process using `kill -9` and its Process ID (PID).

3. Learned Path

The commands across both modules build on one another in a natural, practical sequence:

Step 1 — Orient yourself: Use `pwd` and `cd` to find your current location and move around the filesystem.

Step 2 — List what's there: Use `ls` and its options (`-l`, `-R`, `-a`, `-al`) to inspect directory contents in increasing detail.

Step 3 — Work with file content: Use `cat` to view, create, and copy the contents of text files.

Step 4 — Duplicate and organize: Use `cp` to copy files and `mv` to rename or relocate them.

Step 5 — Clean up: Use `rm` and `rm -r` to remove files and directories that are no longer needed.

Step 6 — Monitor processes: Use `ps -aux` to view running processes and understand what a daemon is.

Step 7 — Manage processes: Use `kill -9 PID` to forcefully terminate a specific process.

4. Navigation

Command	Description
<code>pwd</code>	Displays the current working directory — your current location in the filesystem.

Command	Description
cd	Changes the current directory, allowing you to move between directories.

The screenshot shows a VS Code terminal window with the following commands and output:

```

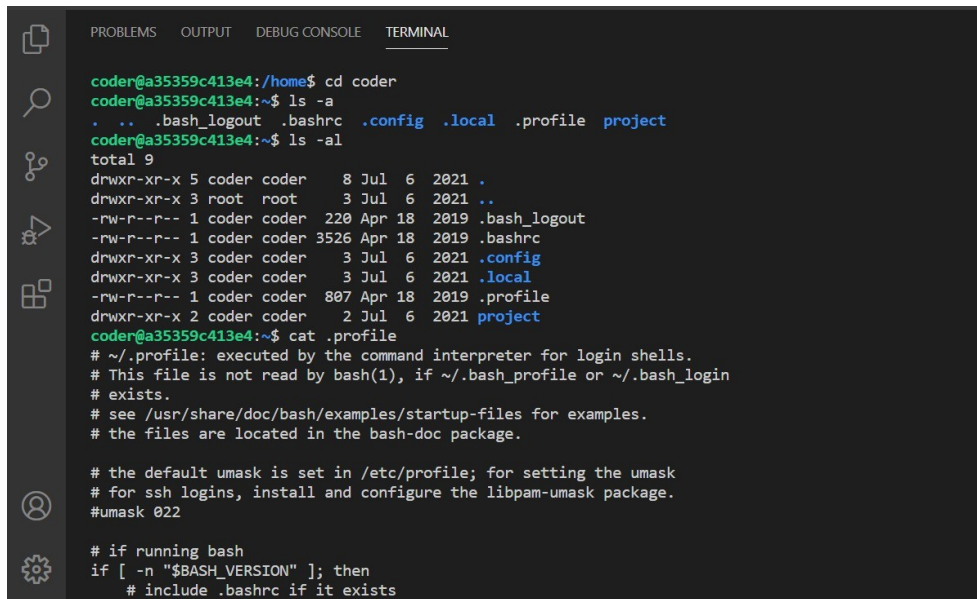
coder@a35359c413e4:~$ pwd
/home/coder
coder@a35359c413e4:~$ cd
coder@a35359c413e4:~$ ls coder
ls: cannot access 'coder': No such file or directory
coder@a35359c413e4:~$ ls
project
coder@a35359c413e4:~$ cd ..
coder@a35359c413e4:/home$ pwd
/home
coder@a35359c413e4:/home$ cd coder
coder@a35359c413e4:/home/coder$ pwd
/home/coder
coder@a35359c413e4:/home/coder$ ls
project
coder@a35359c413e4:/home/coder$ cd ..
coder@a35359c413e4:/home$ ls
coder
coder@a35359c413e4:/home$ ls -R
.:
coder
./coder:
project
./coder/project:
coder@a35359c413e4:/home$ ls -a

```

Figure 1 — Using `pwd` and `cd` to move between directories.

5. Directory & File Listing

Command	Description
ls	Lists the files and directories in the current directory.
ls -l	Displays files and directories in long format, showing permissions, owner, size, and modification date.
ls -R	Recursively lists all files and directories, including the contents of all subdirectories.
ls -a	Lists all files and directories, including hidden files (those beginning with a dot).
ls -al	Combines <code>-l</code> and <code>-a</code> : a detailed listing of all files, including hidden ones.



```
coder@a35359c413e4:/home$ cd coder
coder@a35359c413e4:~$ ls -a
.  ..  .bash_logout  .bashrc  .config  .local  .profile  project
coder@a35359c413e4:~$ ls -al
total 9
drwxr-xr-x 5 coder coder  8 Jul 6 2021 .
drwxr-xr-x 3 root  root   3 Jul 6 2021 ..
-rw-r--r-- 1 coder coder 220 Apr 18 2019 .bash_logout
-rw-r--r-- 1 coder coder 3526 Apr 18 2019 .bashrc
drwxr-xr-x 3 coder coder  3 Jul 6 2021 .config
drwxr-xr-x 3 coder coder  3 Jul 6 2021 .local
-rw-r--r-- 1 coder coder 807 Apr 18 2019 .profile
drwxr-xr-x 2 coder coder  2 Jul 6 2021 project
coder@a35359c413e4:~$ cat .profile
# ~/.profile: executed by the command interpreter for login shells.
# This file is not read by bash(1), if ~/.bash_profile or ~/.bash_login
# exists.
# see /usr/share/doc/bash/examples/startup-files for examples.
# the files are located in the bash-doc package.

# the default umask is set in /etc/profile; for setting the umask
# for ssh logins, install and configure the libpam-umask package.
#umask 022

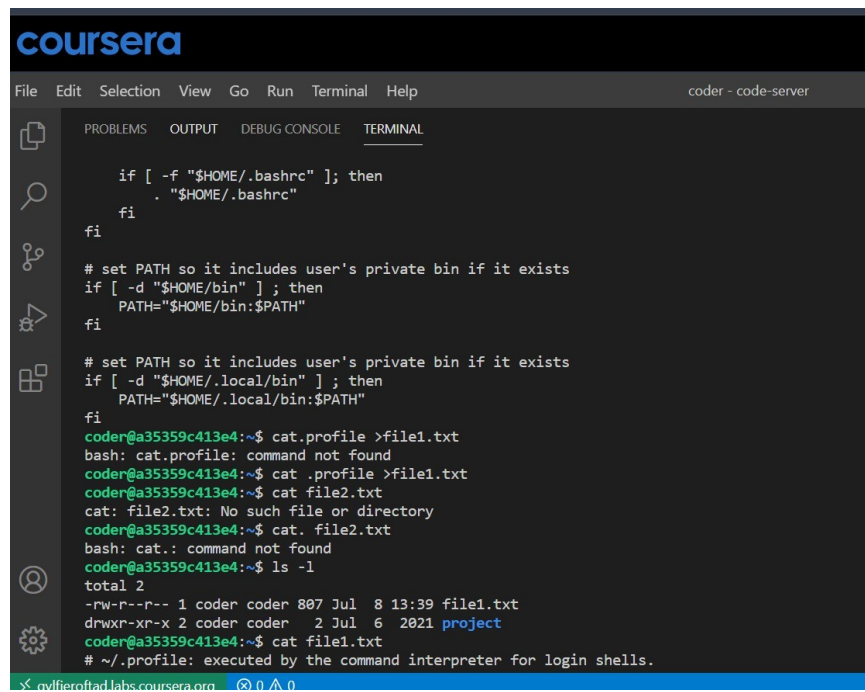
# if running bash
if [ -n "$BASH_VERSION" ]; then
    # include .bashrc if it exists
```

Figure 2 — `ls -a` and `ls -al` listing hidden files in detail.

`ls` and `ls -R` are covered in the table above; `ls` itself is also visible in Figure 1, and `ls -R` was practiced without a dedicated screenshot capture.

6. Viewing & Copying Files

Command	Description
<code>cat</code>	Displays, creates, concatenates (combines), and appends text files.
<code>cat .profile > file</code>	Copies the contents of <code>.profile</code> into <code>file_name</code> , overwriting it if it already exists.
<code>cp</code>	Copies files or directories from one location to another.



```

coursera
File Edit Selection View Go Run Terminal Help
coder - code-server

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

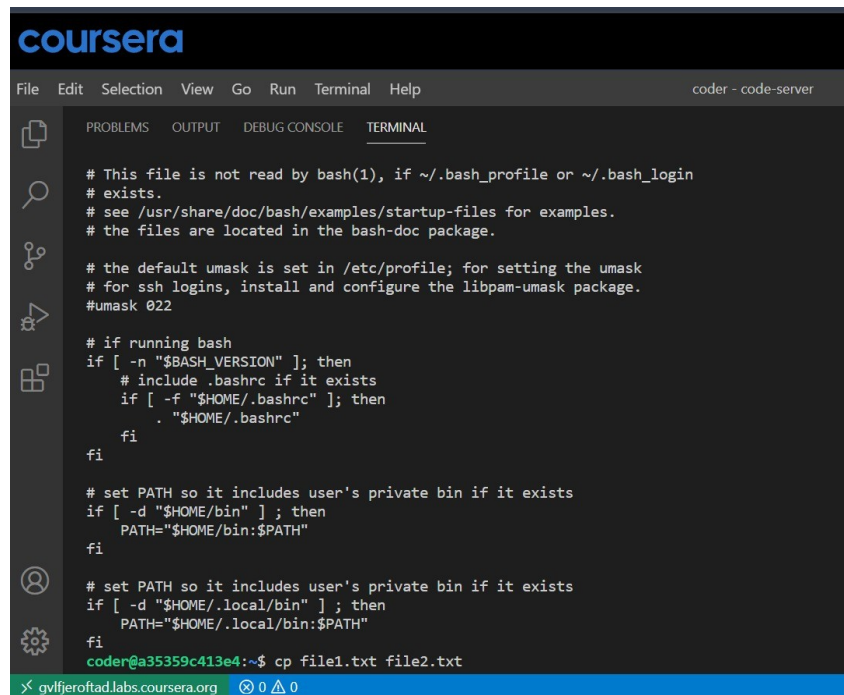
if [ -f "$HOME/.bashrc" ]; then
    . "$HOME/.bashrc"
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/bin" ]; then
    PATH="$HOME/bin:$PATH"
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/.local/bin" ]; then
    PATH="$HOME/.local/bin:$PATH"
fi
coder@a35359c413e4:~$ cat .profile >file1.txt
bash: cat .profile: command not found
coder@a35359c413e4:~$ cat .profile >file1.txt
coder@a35359c413e4:~$ cat file2.txt
cat: file2.txt: No such file or directory
coder@a35359c413e4:~$ cat .file2.txt
bash: cat.: command not found
coder@a35359c413e4:~$ ls -l
total 2
-rw-r--r-- 1 coder coder 807 Jul  8 13:39 file1.txt
drwxr-xr-x 2 coder coder  2 Jul  6 2021 project
coder@a35359c413e4:~$ cat file1.txt
# ~/.profile: executed by the command interpreter for login shells.

```

Figure 3 — Viewing the contents of a startup file with cat.



```

coursera
File Edit Selection View Go Run Terminal Help
coder - code-server

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

# This file is not read by bash(1), if ~/.bash_profile or ~/.bash_login
# exists.
# see /usr/share/doc/bash/examples/startup-files for examples.
# the files are located in the bash-doc package.

# the default umask is set in /etc/profile; for setting the umask
# for ssh logins, install and configure the libpam-umask package.
#umask 022

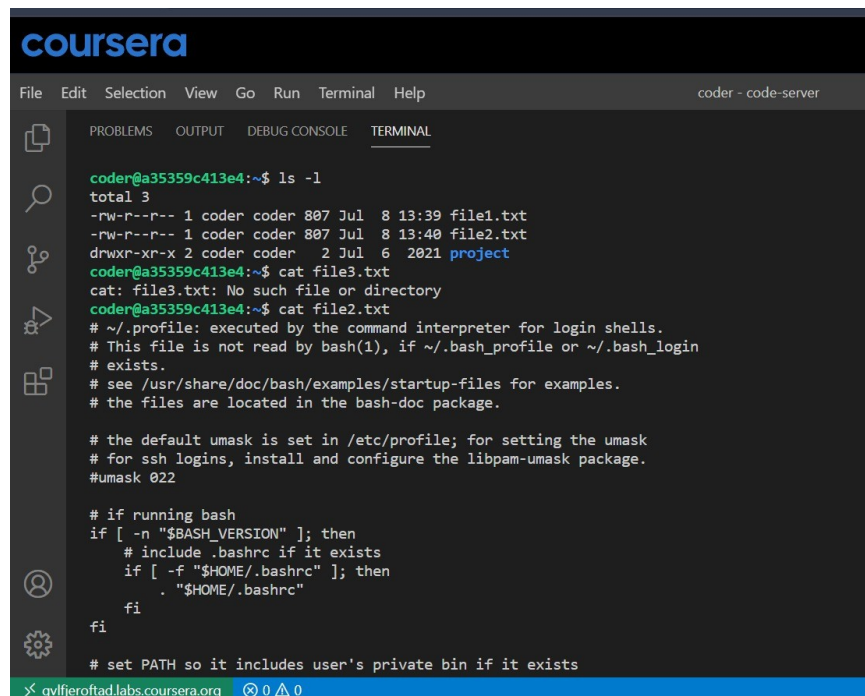
# if running bash
if [ -n "$BASH_VERSION" ]; then
    # include .bashrc if it exists
    if [ -f "$HOME/.bashrc" ]; then
        . "$HOME/.bashrc"
    fi
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/bin" ]; then
    PATH="$HOME/bin:$PATH"
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/.local/bin" ]; then
    PATH="$HOME/.local/bin:$PATH"
fi
coder@a35359c413e4:~$ cp file1.txt file2.txt

```

Figure 4 — Continued file content displayed by cat.

A screenshot of a web-based code editor interface. At the top, the 'coursera' logo is on the left, and a menu bar with 'File', 'Edit', 'Selection', 'View', 'Go', 'Run', 'Terminal', and 'Help' is on the right. Below the menu, there's a toolbar with icons for file operations and a 'TERMINAL' tab. The terminal window shows a shell prompt 'coder@a35359c413e4:~\$' followed by the command 'ls -l'. The output lists three files: 'file1.txt', 'file2.txt', and 'project'. Then, the user runs 'cat file3.txt', which results in an error message: 'cat: file3.txt: No such file or directory'. Finally, the user runs 'cat file2.txt', which displays the contents of the file, including comments about the .profile file, umask settings, and bashrc configuration. The bottom status bar shows the URL 'qvljferoftad.labs.coursera.org' and some icons.

```
coder@a35359c413e4:~$ ls -l
total 3
-rw-r--r-- 1 coder coder 807 Jul  8 13:39 file1.txt
-rw-r--r-- 1 coder coder 807 Jul  8 13:40 file2.txt
drwxr-xr-x 2 coder coder  2 Jul  6 20:21 project
coder@a35359c413e4:~$ cat file3.txt
cat: file3.txt: No such file or directory
coder@a35359c413e4:~$ cat file2.txt
# ~/.profile: executed by the command interpreter for login shells.
# This file is not read by bash(1), if ~/.bash_profile or ~/.bash_login
# exists.
# see /usr/share/doc/bash/examples/startup-files for examples.
# the files are located in the bash-doc package.

# the default umask is set in /etc/profile; for setting the umask
# for ssh logins, install and configure the libpam-umask package.
#umask 022

# if running bash
if [ -n "$BASH_VERSION" ]; then
    # include .bashrc if it exists
    if [ -f "$HOME/.bashrc" ]; then
        . "$HOME/.bashrc"
    fi
fi

# set PATH so it includes user's private bin if it exists
```

Figure 5 — `ls -l` followed by `cat` used on `file2.txt` and a missing `file3.txt`.

`cp` was practiced conceptually in this module; no dedicated terminal screenshot of `cp` was captured.

7. Removing & Renaming Files

Command	Description
rm	Removes (deletes) a file.
rm -r	Removes a directory and all of its contents — files and subdirectories — recursively.
mv	Moves or renames a file or directory.

The screenshot shows a terminal window within a Coursera IDE. The terminal displays a series of commands and their outputs. It starts with a shell script snippet for setting the PATH variable. Then, the user runs `mv file2.txt hamna.txt`. This is followed by `ls -l`, which shows the directory contents, including the newly renamed file `hamna.txt`. The user then runs `ls -l` again, showing the same directory listing. Next, the user runs `mkdir mubashra`, followed by `cd mubashra`. The user then runs `pwd`, which shows the current directory as `/home/coder/mubashra`. The user then runs `cd coder`, which results in an error: `bash: cd: coder: No such file or directory`. The user then runs `cd`, which returns to the `/home/coder/mubashra` directory. The user then runs `rm mubashra`, which results in an error: `rm: cannot remove 'mubashra': Is a directory`. Finally, the user runs `rm -r mubashra`, which successfully removes the directory. The terminal ends with `ls -l`, showing the directory contents after the removal.

```

if [ -d "$HOME/bin" ]; then
  PATH="$HOME/bin:$PATH"
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/.local/bin" ]; then
  PATH="$HOME/.local/bin:$PATH"
fi

coder@a35359c413e4:~$ mv file2.txt hamna.txt
coder@a35359c413e4:~$ ls -l
bash: ls-l: command not found
coder@a35359c413e4:~$ ls -l
total 4
-rw-r--r-- 1 coder coder 807 Jul  8 13:39 file1.txt
-rw-r--r-- 1 coder coder 807 Jul  8 13:40 hamna.txt
drwxr-xr-x 2 coder coder  2 Jul  6 2021 project
coder@a35359c413e4:~$ mkdir mubashra
coder@a35359c413e4:~$ cd mubashra
coder@a35359c413e4:~/mubashra$ pwd
/home/coder/mubashra
coder@a35359c413e4:~/mubashra$ cd coder
bash: cd: coder: No such file or directory
coder@a35359c413e4:~/mubashra$ cd
coder@a35359c413e4:~$ rm mubashra
rm: cannot remove 'mubashra': Is a directory
coder@a35359c413e4:~$ rm -r mubashra
coder@a35359c413e4:~$ ls -l

```

Figure 6 — Renaming a file with `mv` (`file2.txt` → `hamna.txt`).

`rm` and `rm -r` were practiced in this module; no dedicated terminal screenshot of `rm` was captured.

8. Process Management

Command	Description
ps -aux	Displays detailed information about all running processes: user, PID, CPU usage, memory usage, and the command that started each process.
kill -9 PID	Forcefully terminates the process with the specified Process ID (PID).

This module introduces the concept of a daemon (d) — a service or process that runs continuously in the background without requiring user interaction. Many Linux daemon names end in d, such as `mysqld`, `sshd`, and `httpd`. For example, when `ps aux` is run, the process `mysqld` may appear with its own Process ID (PID) — in this session, PID 149 — showing that the MySQL service is running in the background.

The screenshot shows a terminal window titled 'code-server' with a menu bar (File, Edit, Selection, View, Go, Run, Terminal, Help) and tabs (PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, SQL CONSOLE). The terminal prompt is 'coder@81c289dd9af7:~\$'. The first command executed is 'ps -aux', which lists running processes. The second command is 'kill -9 149', which terminates the process with PID 149. The third command is 'ps -aux', which shows the updated list of processes, with PID 149 no longer present.

```
coder@81c289dd9af7:~$ ps -aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
coder           1  0.0  0.0   2368    256 ?        Ss   14:21   0:00 dumb-init -- start.sh
coder           7  0.0  0.0   6900   1280 ?        Ss   14:21   0:00 bash /usr/local/bin/start.sh
coder          10  0.4  0.0  611000 32408 ?        S1   14:21   0:00 /usr/lib/code-server/lib/node /usr/lib/code-server --config /tmp/config/code-server/config.yaml
coder          44  0.4  0.0  613164 34912 ?        S1   14:21   0:00 /usr/lib/code-server/lib/node /usr/lib/code-server --config /tmp/config/code-server/config.yaml
coder          77  1.1  0.0  1067664 45060 ?        R1   14:21   0:01 /usr/lib/code-server/lib/node /usr/lib/code-server/lib/vscode/out/vs/server/fork
coder         111  1.3  0.0  1060104 78452 ?        S1   14:21   0:01 /usr/lib/code-server/lib/node /usr/lib/code-server/lib/vscode/out/bootstrap-fork --type=extensi
coder         126  0.2  0.0  587880 40164 ?        S1   14:22   0:00 /usr/lib/code-server/lib/node /home/coder/.local/share/code-server/extensions/mtxr.sqltools-0.2
coder         141  0.0  0.0    7244   2304 pts/0    Ss   14:22   0:00 /bin/bash
coder         149  1.2  0.1 1937596 328452 pts/0    Sl+  14:23   0:00 mysql
coder         193  0.0  0.0    7244   2304 pts/1    Ss   14:23   0:00 /bin/bash
coder         200  0.0  0.0    9140   1792 pts/1    R+   14:24   0:00 ps -aux
coder@81c289dd9af7:~$ kill -9 149
coder@81c289dd9af7:~$ ps -aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
coder           1  0.0  0.0   2368    256 ?        Ss   14:21   0:00 dumb-init -- start.sh
coder           7  0.0  0.0   6900   1280 ?        Ss   14:21   0:00 bash /usr/local/bin/start.sh
coder          10  0.2  0.0  611000 32408 ?        S1   14:21   0:00 /usr/lib/code-server/lib/node /usr/lib/code-server --config /tmp/config/code-server/config.yaml
coder          44  0.3  0.0  613164 34912 ?        S1   14:21   0:00 /usr/lib/code-server/lib/node /usr/lib/code-server --config /tmp/config/code-server/config.yaml
coder          77  0.9  0.0  1067664 44624 ?        R1   14:21   0:01 /usr/lib/code-server/lib/node /usr/lib/code-server/lib/vscode/out/vs/server/fork
coder         111  1.0  0.0  1060104 78964 ?        S1   14:21   0:02 /usr/lib/code-server/lib/node /usr/lib/code-server/lib/vscode/out/bootstrap-fork --type=extensi
coder         126  0.2  0.0  587880 40164 ?        S1   14:22   0:00 /usr/lib/code-server/lib/node /home/coder/.local/share/code-server/extensions/mtxr.sqltools-0.2
coder         141  0.0  0.0    7244   2304 pts/0    Ss+  14:22   0:00 /bin/bash
coder         193  0.0  0.0    7244   2304 pts/1    Ss   14:23   0:00 /bin/bash
coder         201  0.0  0.0    9140   2048 pts/1    R+   14:25   0:00 ps -aux
```

Figure 7 — `ps -aux` listing running processes, then `kill -9` terminating the `mysql` process.

9. Summary

Together, Modules 1 and 2 build a practical foundation for working in the Linux command-line environment: orienting yourself in the filesystem, listing and reading files in detail, copying and reorganizing them, cleaning up what's no longer needed, and finally looking beyond files to the processes — including background daemons — that keep a Linux system running.